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Editorial

Clinical Simulation in Medicine: Past, Present, and, Above All, the Future

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Simulación clínica en medicina: pasado, presente y, sobre todo, el futuro

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Clinical simulation has traditionally been defined as an educational technique designed to replace real-life experiences with guided experiences that evoke or replicate aspects of the real world in an interactive manner [1]. This definition, proposed by Gaba nearly two decades ago, remains of fundamental importance because it emphasizes that simulation is a technique rather than a technology. Consequently, access to high-fidelity simulators or state-of-the-art equipment is not a prerequisite for delivering high-quality clinical simulation.

The quality of clinical simulation lies in its instructional design, clearly defined educational objectives, and, above all, the quality of the feedback process following the simulated clinical scenario (debriefing).

In one of the chapters of the Manual de Simulación Clínica en Especialidades Médicas by Barroso-González and

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Herrera-Pérez, the authors pose several questions that they consider relevant when defining simulation and that clearly illustrate the concept that simulation is fundamentally a technique rather than a technology [2]

- Can the practice introduced in the third century BC by the Indian physician Sushruta, who used pieces of fruit to teach students how to perform surgical incisions, be regarded as simulation?
- Likewise, should the use of animal models and cadavers for surgical training, practiced since antiquity as a means of acquiring anatomical knowledge, also be considered simulation?

In my view, the answer is unequivocally yes. Clinical simulation should be understood as any educational activity aimed at acquiring new technical or non-technical skills without practicing on real patients, thereby challenging the traditional "see one, do one, teach one" paradigm of medical education.

The surgeon William Stewart Halsted at Johns Hopkins University popularized this educational model in the late nineteenth century [3]. There is no doubt that it has trained thousands of physicians over several decades and continues—and will continue—to play a highly significant role in clinical education. Nevertheless, the increasing complexity of modern medicine, together with growing demands for healthcare quality, ethical standards, and patient safety, has required a reassessment of traditional educational approaches.

Today, it is difficult to justify a healthcare student's or resident physician's first experience with an invasive procedure taking place on a real patient. One of the greatest advantages of clinical simulation—arguably, it is most important—is that it enables learners to develop competencies within a safe environment, thereby contributing directly to improved patient safety.

Clinical simulation is essential for enhancing patient safety, particularly in high-stress situations in which learning must focus not only on technical skills (such as effective airway management) but also on non-technical skills, including communication, situational awareness, leadership, and teamwork [4].

Historically, non-technical skills have received considerably less attention than technical skills despite being essential for safe, high-quality clinical practice [5]. Clinical simulation provides an ideal environment in which to develop and reinforce these competencies. Therefore, all healthcare professionals—including physicians, medical students, nurses, nursing assistants, and healthcare support staff—should be actively involved in simulation-based training.

These competencies should be cultivated from the earliest stages of undergraduate medical education, and clinical simulation should become a cornerstone in the training of future physicians and healthcare professionals. Across medical school curricula, simulation-based education is expanding rapidly and irreversibly, both in long-established medical schools with centuries of tradition and in newly established medical programs [6]. Its importance is also increasingly recognized within postgraduate specialty training. For example, the recent revision of the Intensive Care Medicine training curriculum explicitly states that clinical simulation should play a role in the assessment of medical residents [7].

Finally, I would like to express my gratitude for the opportunity to write this editorial and encourage all healthcare professionals to embrace clinical simulation and incorporate it into both their educational and clinical practice, regardless of their specialty or level of experience. Clinical simulation represents an investment in education but, above all, an investment in patient safety. Ultimately, our patients will be the greatest beneficiaries.

1. CONFLICT OF INTERESTS

The authors have no conflict of interest to declare. The authors declared that this study has received no financial support.

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